Lecture 20: Evaluating LLMs

Instructor: Stephanie Schoch

CSCI 680: Natural Language Processing
Department of Computer Science, College of William & Mary
November 4, 2026

WILLIAM & MARY

Some slides adapted from Jurafsky & Martin, Chris Manning, Swabha Swayamdipta, and Yejin Choi

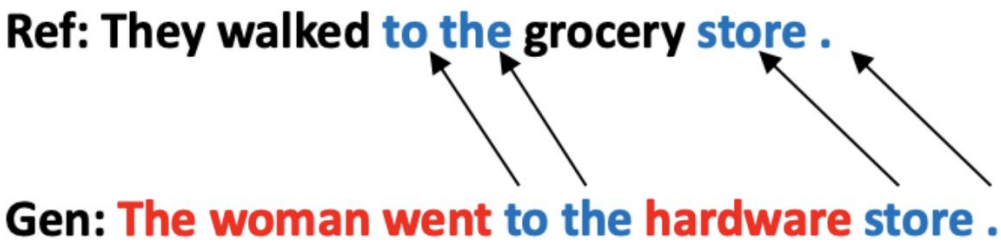


Lecture Outline

- Evaluation Metrics: “How to evaluate?”
  - Automated Metrics
    - Classical Metrics (e.g. overlap metrics)
    - Learned metrics: (e.g. BERTScore)
  - Human Evaluation
  - LLM-as-a-Judge
- Benchmarks: “What to evaluate on?”
  - Limitations (e.g. benchmark saturation)

Metrics: “How to evaluate”

Classical, Overlap-based Metrics



- Compute a score that indicates the lexical similarity between generated and gold-standard (human-written) text
- Fast and efficient and widely used
- *n*-gram overlap metrics (e.g., BLEU, ROUGE, etc.)

BLEU

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

$$p_n = \frac{\sum_{\text{n-gram} \in c} \min\left(\text{Count}(\text{n-gram}, c),\ \text{Count}(\text{n-gram}, \text{ref})\right)}{\sum_{\text{n-gram} \in c} \text{Count}(\text{n-gram}, c)}$$

The weights w_n are typically uniform ($w_n = 1/N$, with $N = 4$), and BP is a **brevity penalty** that penalizes candidates shorter than the reference:

$$\text{BP} = \begin{cases} 1 & \text{if } |c| > |r| \\ \exp\left(1 - |r|/|c|\right) & \text{if } |c| \leq |r| \end{cases}$$

where $|c|$ and $|r|$ are the lengths of the candidate and reference.

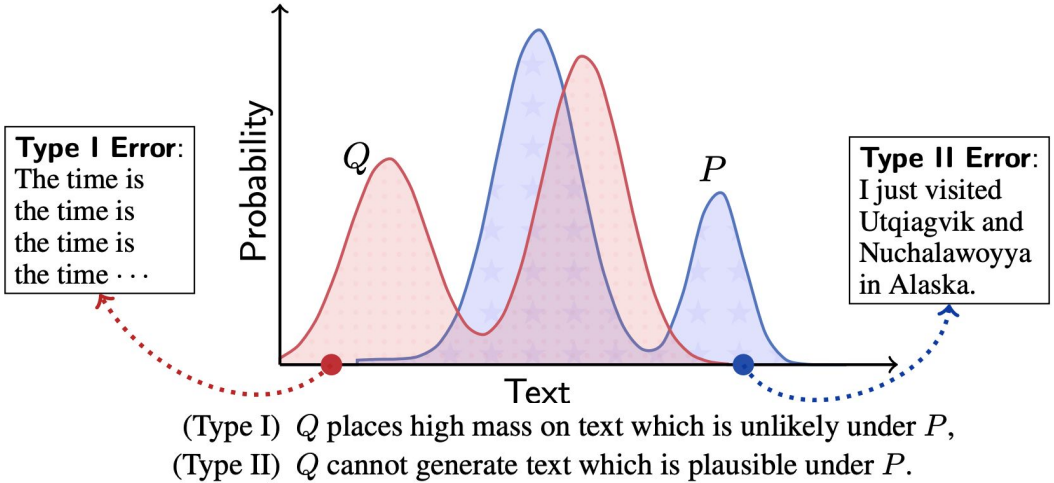
- BLEU: n-gram count-based metric to capture the similarity between the reference (gold-standard, usually human-written) text and the model generated output
- Too rigid on exact surface patterns, not semantically meaningful enough!
- Because BLEU is a word-based metric, it is very sensitive to word tokenization, making it impossible to compare different systems if they rely on different tokenization

[Papineni et al., 2002]

MAUVE: Measuring the Gap Between Neural Text and Human Text

[Pillutla et al., 2021]

Repetitive, degenerate text.



Plausible human text

- Compare human / natural language distributions to model-generated language distributions
- Divergence between these two distributions can be measured by MAUVE

Evaluating Generation: Other Options

- Perplexity
- ROUGE (recall-oriented metric between candidate and set of references)
- Diversity metic (e.g Distinct-N)
- Model-based Metrics (BERTScore, BARTScore, Word Mover's Distance, BLEURT)
  - Use learned representations of words and sentences to compute semantic similarity between generated and reference texts
  - No more n-gram bottleneck because text units are represented as embeddings!
  - The embeddings are pretrained, distance metrics used to measure the similarity can be fixed
- Automatic metrics for text comparisons fall short of matching human decisions
- So, Human Evaluation!

Metrics, Metrics, Metrics

Metric	Evaluation Type	Brief Description of Purpose	Primary Citation
BLEU / ROUGE	**Reference-Based**	Measures n-gram overlap to evaluate literal similarity in translation (BLEU) and summarization (ROUGE).	**Papineni et al. (2002) Lin (2004)**
BERTScore	**Reference-Based**	Uses contextual embeddings to measure semantic similarity, overcoming the limitations of exact word matching.	**Zhang et al. (2020)**
COMET	**Reference-Based**	Neural metrics that use source and reference text to predict human-like quality scores and detect fine-grained errors.	**Rei et al. (2020)**
COMETKiwi / QE	**Reference-Free**	A "Quality Estimation" metric that evaluates translations by comparing them directly to the source without a reference.	**Rei et al. (2022)**
FActScore	**Reference-Free**	Breaks long-form text into atomic claims and verifies each against a knowledge source like Wikipedia.	**Min et al. (2023)**
CLIPScore	**Reference-Free**	Evaluates image captions by calculating the visual-textual alignment between pixels and generated text.	**Hessel et al. (2021)**
SelfCheckGPT	**Reference-Free**	Detects hallucinations by measuring the consistency of facts across multiple internal samples from the same model.	**Manakul et al. (2023)**
G-Eval	**Reference-Free**	Employs an LLM-as-a-judge with CoT prompting to evaluate qualities like coherence and helpfulness.	**Liu et al. (2023)**

Human Evaluation → LLM-as-a-Judge Evaluation

Human Evaluation

- Ask humans to evaluate the quality of generated text
  - Along specific axes: fluency, coherence / consistency, factuality and correctness, commonsense, etc.
- Mostly done via crowdsourcing
- Human judgments are still regarded as the gold standard
- **But human judgments are tricky!**

Criterion	Total	Criterion	Total
Fluency	13	Manipulation check	3
Naturalness	8	Informativeness	3
Quality	5	Correctness	3
Meaning preservation	5	Syntactic correctness	2
Relevance	5	Qualitative analysis	2
Grammaticality	5	Appropriateness	2
Overall quality	4	Non-redundancy	2
Readability	4	Semantic adequacy	2
Clarity	3	Other criteria	25

Scale	Count
Likert (5-point)	14
Preference	10
Likert (2-point)	6
Likert (3-point)	5
Other Likert (4,7,10-point)	5
Rank-based Magnitude Estimation	5
Free text comments	1

[van der Lee et al., 2019]

Human Evaluation: Limitations

[Clark et al., 2021]

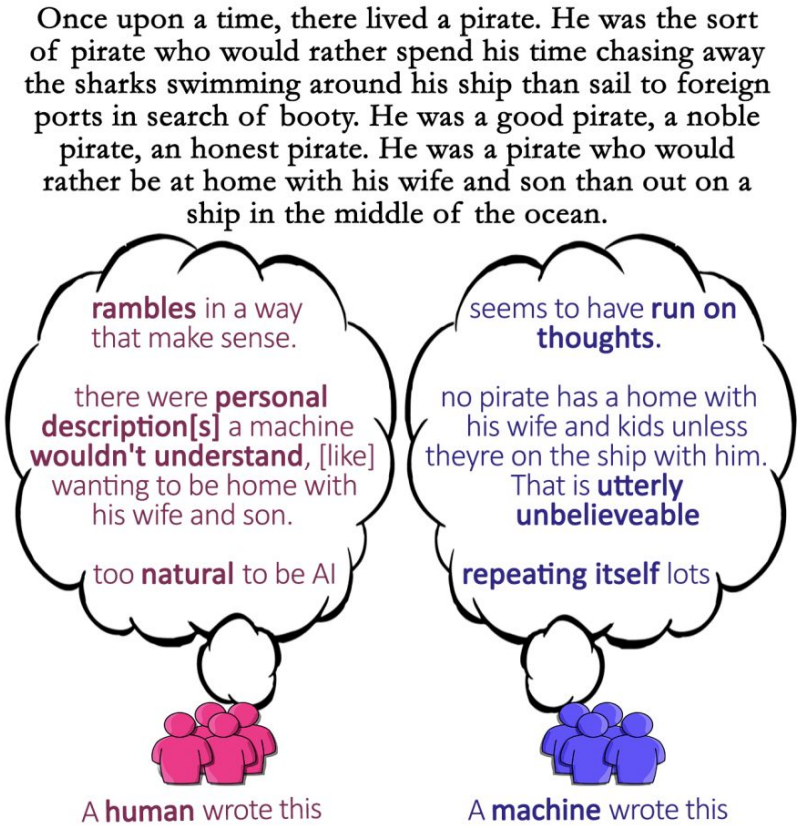
- Reproducibility issues…
  - Wording
  - Formatting
  - Etc.
- Annotation guidelines can be tricky to design
- Biases & heuristics
- Different preferences or thought processes = limited consistency
  - **Rethinking disagreement?**
- Rethinking human evaluation: [Elangovan et al., 2026]

Once upon a time, there lived a pirate. He was the sort of pirate who would rather spend his time chasing away the sharks swimming around his ship than sail to foreign ports in search of booty. He was a good pirate, a noble pirate, an honest pirate. He was a pirate who would rather be at home with his wife and son than out on a ship in the middle of the ocean.

Human Evaluation: Limitations

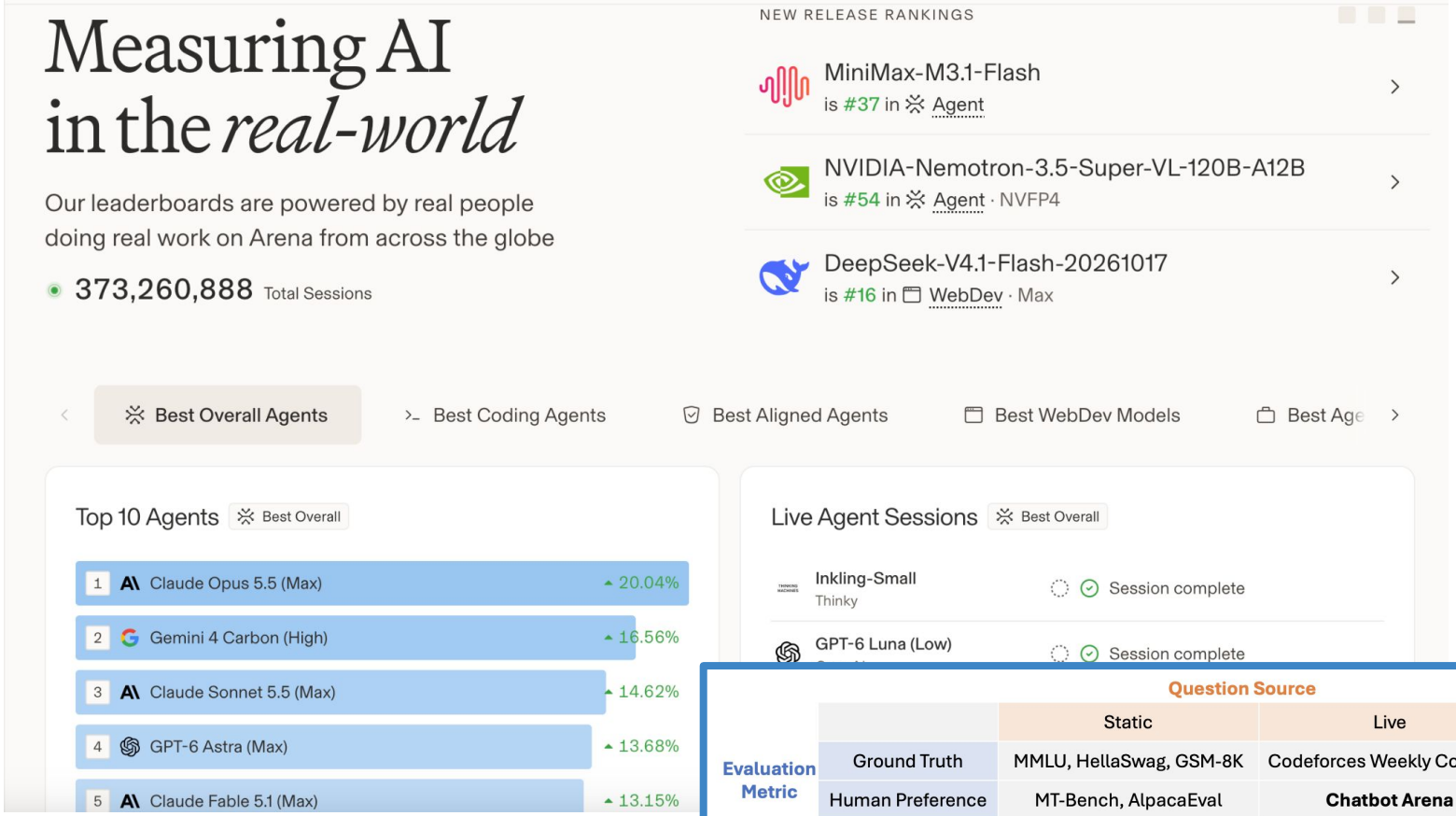
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Chatbot Arena

Link



Evaluation Metric	Question Source: Static	Question Source: Live
Ground Truth	MMLU, HellaSwag, GSM-8K	Codeforces Weekly Contests
Human Preference	MT-Bench, AlpacaEval	**Chatbot Arena**

Chatbot Arena

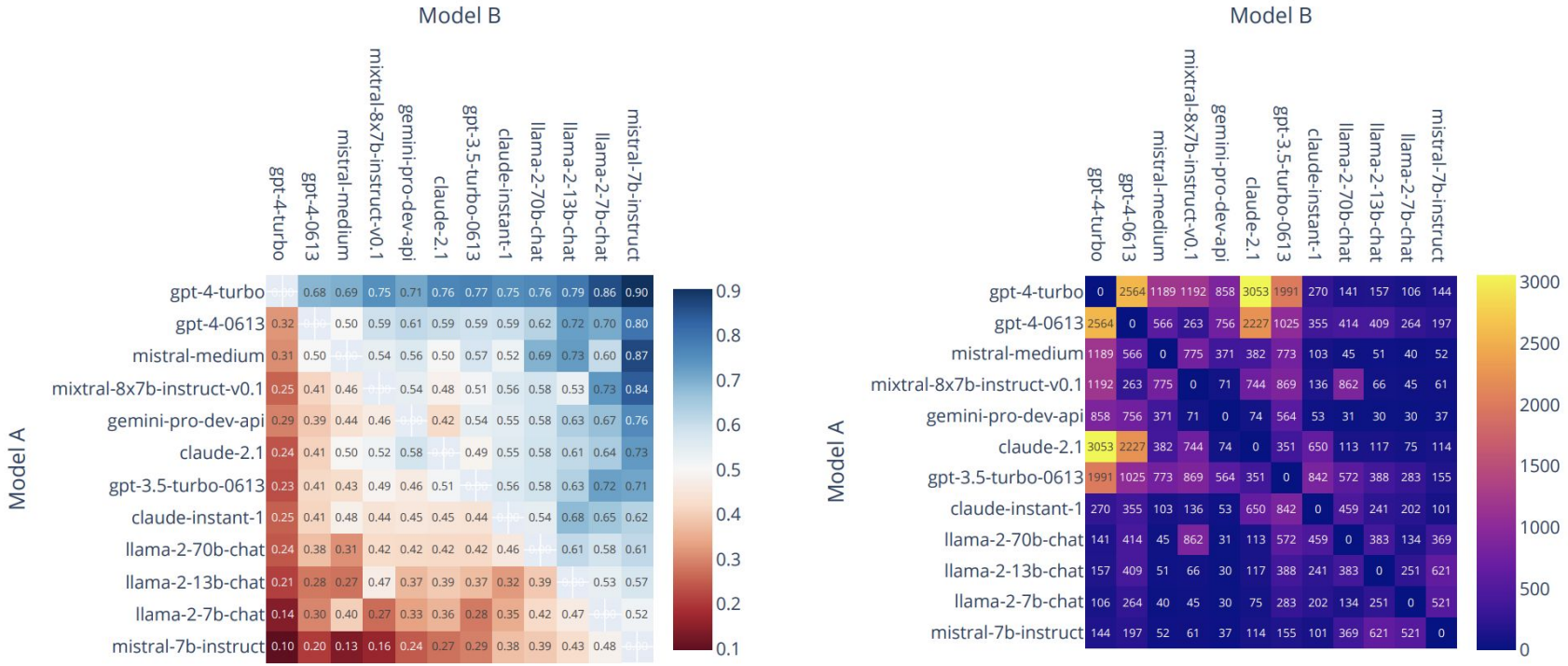
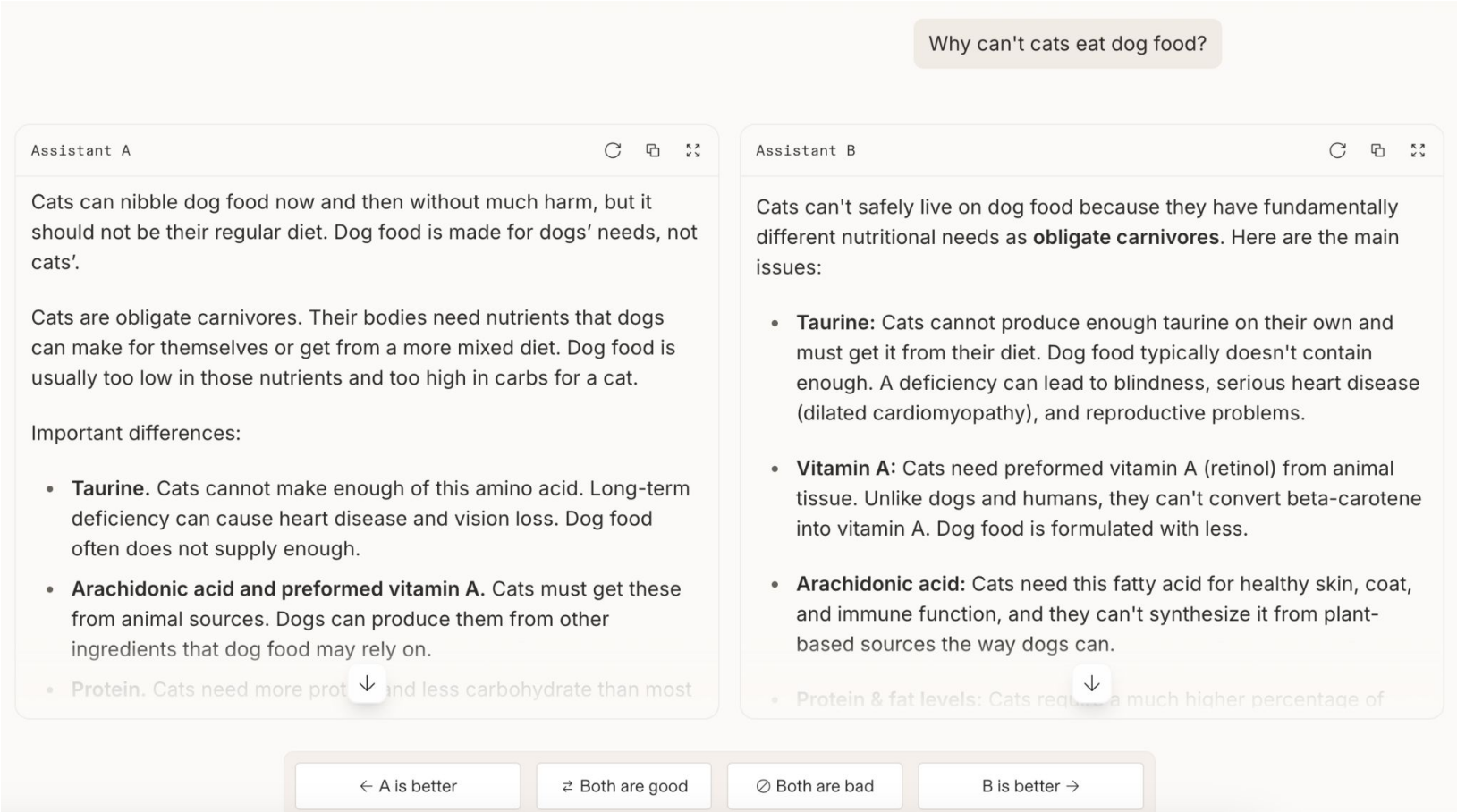


Figure 2. Win-rate (left) and battle count (right) between a subset of models in Chatbot Arena.

Chatbot Arena



Chatbot Arena

Why can't cats eat dog food?

grok-4.7-xhigh

Important differences:

- **Taurine.** Cats cannot make enough of this amino acid. Long-term deficiency can cause heart disease and vision loss. Dog food often does not supply enough.
- **Arachidonic acid and preformed vitamin A.** Cats must get these from animal sources. Dogs can produce them from other ingredients that dog food may rely on.
- **Protein.** Cats need more protein and less carbohydrate than most dog foods provide.
- **Other vitamins** (such as niacin). Cats have higher or different requirements that dog food is not balanced to meet.

Fed only dog food over time, a cat can develop serious nutritional deficiencies. A few bites from the dog's bowl are usually fine; their everyday food should be complete cat food.

qwen3.8-27b

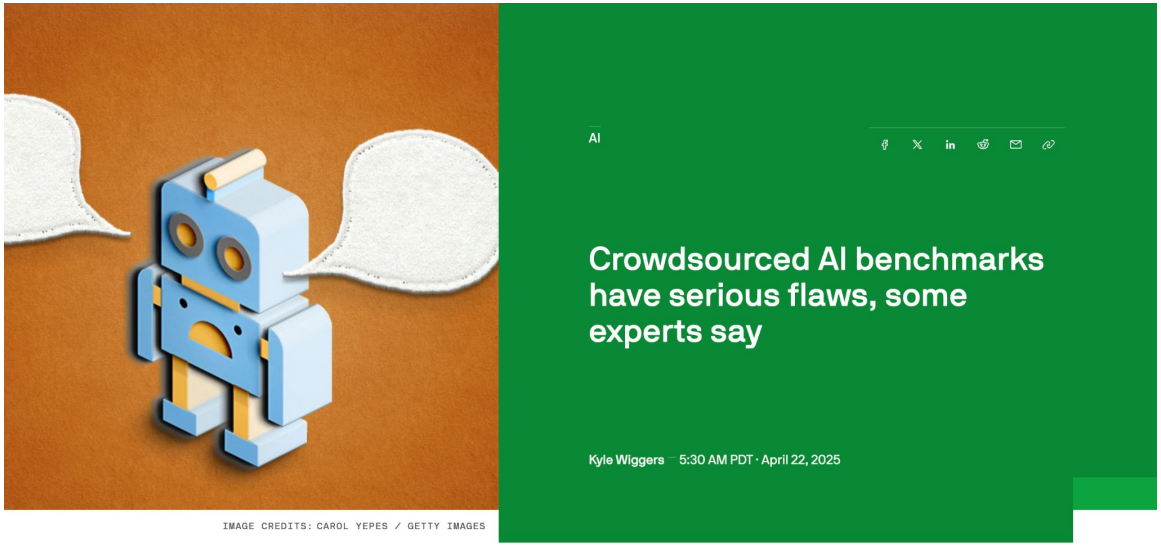
based sources the way dogs can.

- **Protein & fat levels:** Cats require a much higher percentage of animal-based protein and fat in their diet than dogs do. Dog food is formulated for a more omnivorous animal, so it's lower in protein and higher in carbohydrates.
- **Other micronutrients:** The levels of vitamins, minerals, and amino acids in dog food simply don't match what a cat's body requires.

A cat eating dog food for a short time probably won't be in immediate danger, but over weeks or months it can develop taurine deficiency, vitamin A deficiency, and other nutritional problems that are costly or irreversible to fix.

The short version: dog food is the "wrong recipe" for a cat's biology, and cats don't have the metabolic flexibility to compensate the way dogs do.

Chatbot Arena



Potential issues?

- Bias and lack of transparency
- Self-selection of users: they're interested in model evaluation
- User ability to spot hallucinations not tested
- Preferences not well-defined and may differ
- Commercial ties

https://techcrunch.com/2025/04/22/crowdsourced-ai-benchmarks-have-serious-flaws-some-experts-say/

LLM-as-a-Judge

- Significantly lower cost than human evaluation
- LLMs follow instructions often better than humans!
- But even LLMs have flaws…
  - Self-preference / Nepotism bias (over humans' output, or over other models' output)
  - Verbosity bias
  - Weaker at subtle logical flaws
  - Stronger models are costly



Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Lianmin Zheng[1*] Wei-Lin Chiang[1*] Ying Sheng[4*] Siyuan Zhuang[1]

Zhanghao Wu[1] Yonghao Zhuang[3] Zi Lin[2] Zhuohan Li[1] Dacheng Li[13]

Eric P. Xing[35] Hao Zhang[12] Joseph E. Gonzalez[1] Ion Stoica[1]

[1] UC Berkeley [2] UC San Diego [3] Carnegie Mellon University [4] Stanford [5] MBZUAI

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What helps?

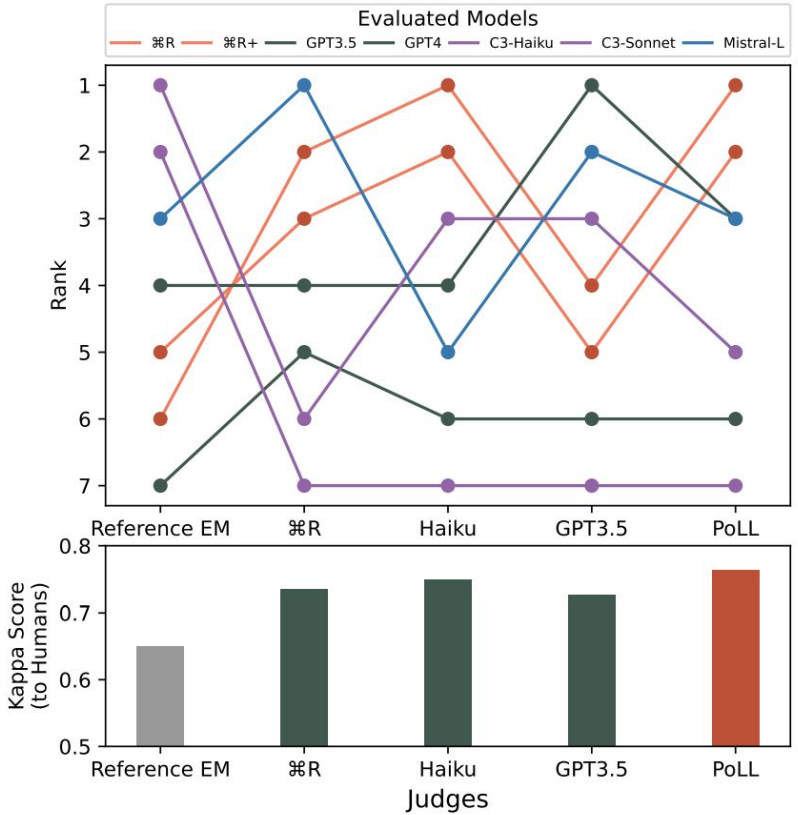
- Clear instructions, examples, rubrics!
- Let them discuss!
  - ChatEval (Chan et al., 2023)
  - CollabEval (Qian et al., 2025)

LLM Juries

Replacing Judges with Juries:
Evaluating LLM Generations with a Panel of Diverse Models

Pat Verga
Sebastian Hofstätter, Sophia Althammer, Yixuan Su
Aleksandra Piktus, Arkady Arkhangorodsky, Minjie Xu, Naomi White
Patrick Lewis
Cohere

- "LLMs as Juries"
  - Bias mitigation via a panel of diverse evaluators
  - Robustness of aggregated scores
  - Cost efficiency by leveraging smaller models

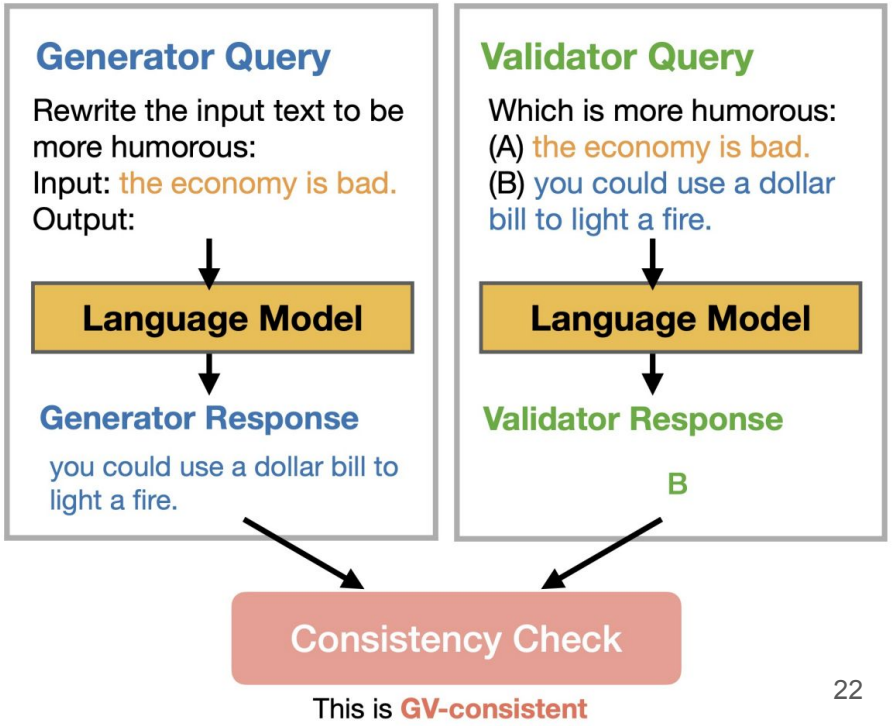


Generator-Validator Gap

THE GENERATIVE AI PARADOX:
"What It Can Create, It May Not Understand"

BENCHMARKING AND IMPROVING
GENERATOR-VALIDATOR CONSISTENCY OF LMS

- "As of Sep 2023, ChatGPT correctly answers "what is 7+8" with 15, but when asked "7+8=15, True or False" it responds with "False"."
- "Even GPT-4, a state-of-the-art LM, is GV-consistent only 76% of the time…"

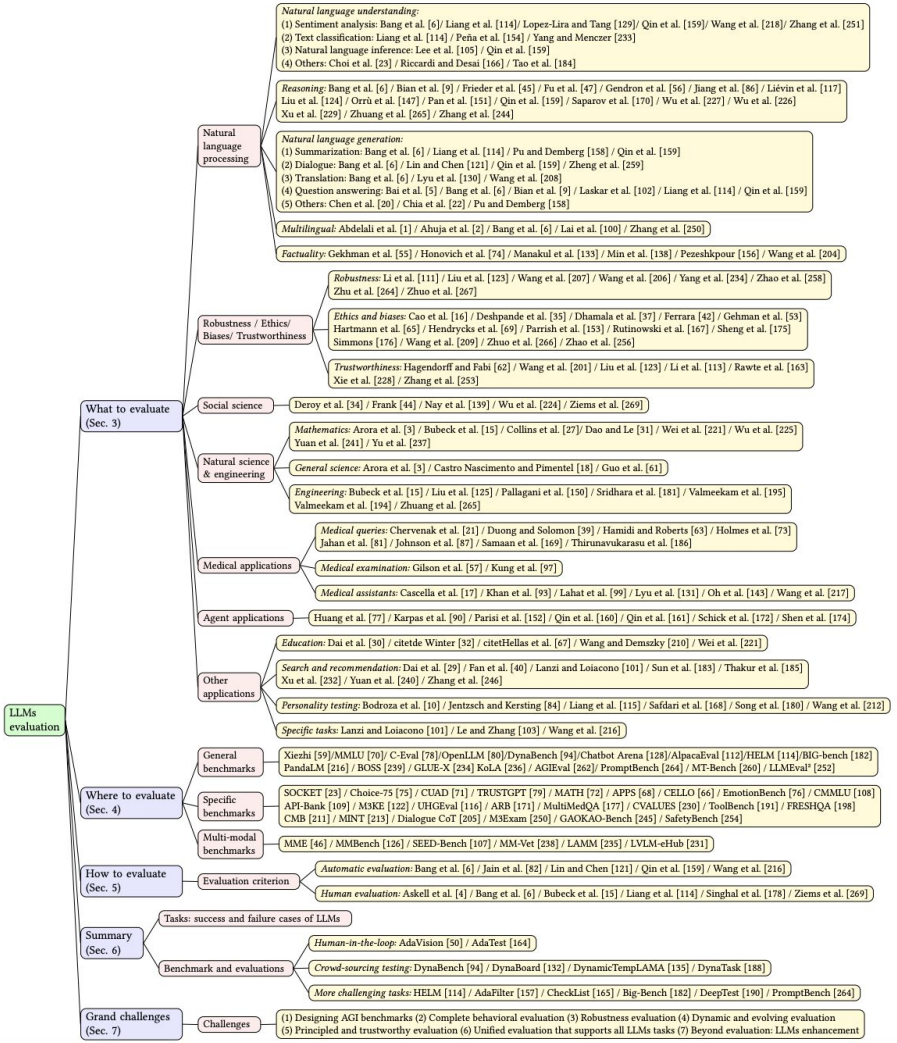


Benchmarks

LLM Evaluation Places A Lot of Emphasis on New Benchmarks

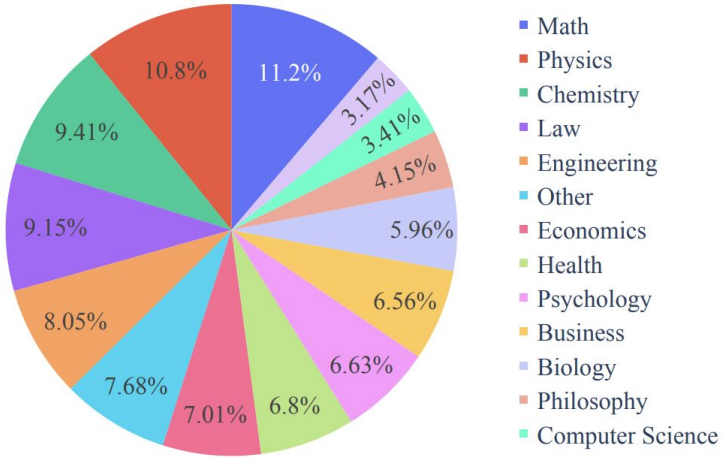
[A Survey on Evaluation of Large Language Models]

Benchmark: a standardized framework for measuring and comparing LLM performance



MMLU

- Massive Multi-Task Language Understanding [Hendrycks et al., 2021]
- 57 tasks, multiple-choice, collected online
- One of the most commonly used benchmarks for measuring LLM performance
- Often used for performance tracking during pre-training (as well as during post-training)
- MMLU-Pro [Wang et al., 2024]:
  - Removed noisy/trivial questions from MMLU
  - Expanded 4 choices to 10 choices



(a) Distribution of Disciplines in MMLU-Pro

MMLU Examples

Astronomy

What is true for a type-Ia supernova?

A. This type occurs in binary systems.
B. This type occurs in young galaxies.
C. This type produces gamma-ray bursts.
D. This type produces high amounts of X-rays.

Answer: A

High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

A. directional selection.
B. stabilizing selection.
C. sexual selection.
D. disruptive selection

Answer: A

Desiderata of Good Benchmarks

- Scale and diversity
  - Benchmark should cover the phenomena of interest
  - Complex phenomena require many and diverse samples
- Difficulty
  - Easy enough for humans (or human experts)
  - Hard enough for state-of-the-art
- Quality
  - Correct answers should be clearly correct
    - Surprisingly hard to guarantee this without a lot of effort!
  - **No spurious bias**
    - Aka “Annotation artifacts”
    - Otherwise, AI can solve exams right for the wrong reasons!

Spurious Bias

[Gururangan et al., 2018]

Lexical overlap, position bias, **annotation artifacts**…

Reminder about NLI:

Entailment	h is definitely true given p
Neutral	h might be true given p
Contradiction	h is definitely **not** true given p

Annotation guidelines example:

Premise	Two dogs are running through a field.
Entailment	There are **animals outdoors**.
Neutral	Some puppies are running **to catch a stick**.
Contradiction	The pets are **sitting on a couch**.

Example SNLI instance:

Premise	A woman selling bamboo sticks talking to two men on a loading dock.
Entailment	There are **at least** three **people** on a loading dock.
Neutral	A woman is selling bamboo sticks **to help provide for her family.**
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Common heuristics:

- **Entailment:** remove gender or number
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Premise oblivious classifier:

Model	SNLI	MultiNLI	
		Matched	Mismatched
majority class	34.3	35.4	35.2
`fastText`	**67.0**	**53.9**	**52.3**

Right for the Wrong Reasons: Diagnosis Syntactic Heuristics in NLI

What if our model is using simple heuristics to get good accuracy?

- A **diagnostic test set** is carefully constructed to test for a specific skill or capacity of your neural model.
- For example, HANS: (Heuristic Analysis for NLI Systems) tests syntactic heuristics in NLI
- **But**, you have to have some notion of the types of problems that may exist…

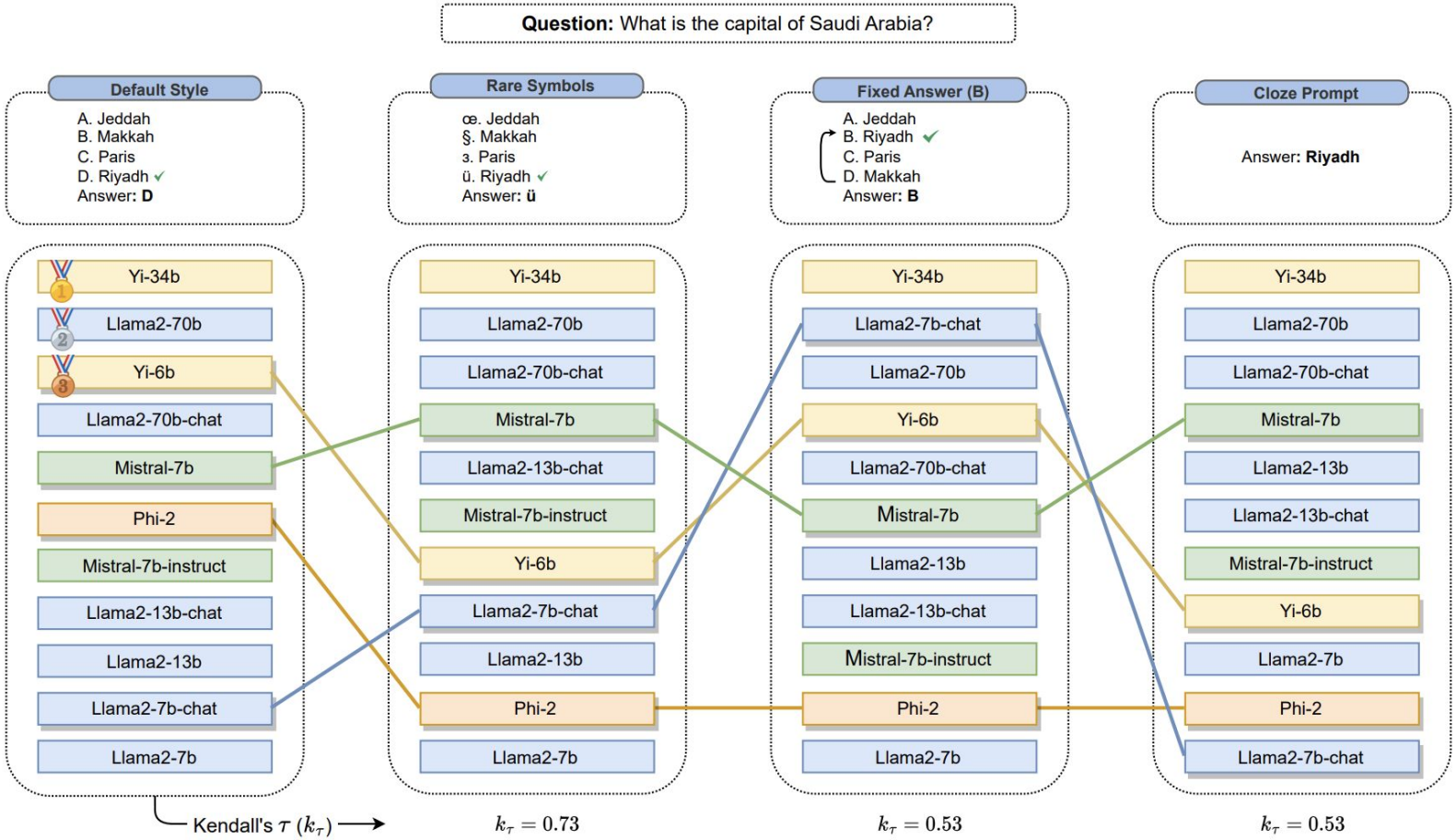
Heuristic	Definition	Example
Lexical overlap	Assume that a premise entails all hypotheses constructed from words in the premise	**The doctor** was **paid** by **the actor**. $\xrightarrow[\text{WRONG}]{}$ The doctor paid the actor.
Subsequence	Assume that a premise entails all of its contiguous subsequences.	The doctor near **the actor danced**. $\xrightarrow[\text{WRONG}]{}$ The actor danced.
Constituent	Assume that a premise entails all complete subtrees in its parse tree.	If **the artist slept**, the actor ran. $\xrightarrow[\text{WRONG}]{}$ The artist slept.

Table 1: The heuristics targeted by the HANS dataset, along with examples of incorrect entailment predictions that these heuristics would lead to.

[McCoy et al., 2019]

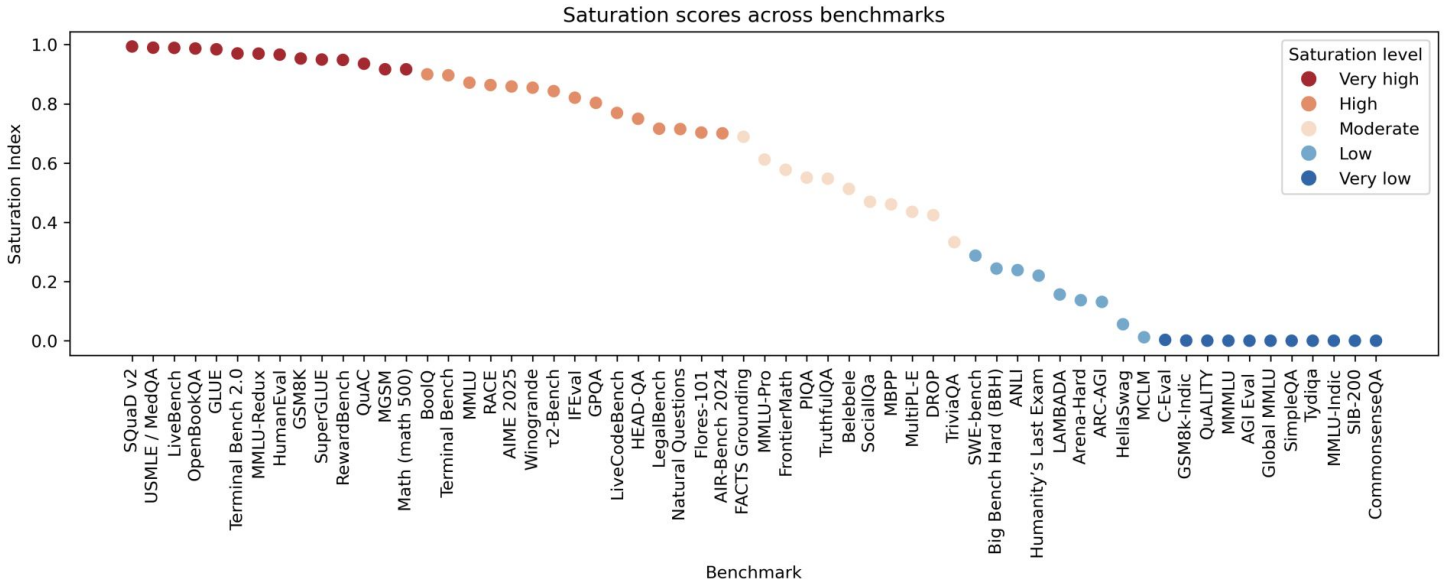
When Benchmarks are Targets: Revealing the Sensitivity of LLM Leaderboards

Alzahrani et al., 2024



- MMLU
- Rankings change up to 8 positions
- Lower τ = more disagreement

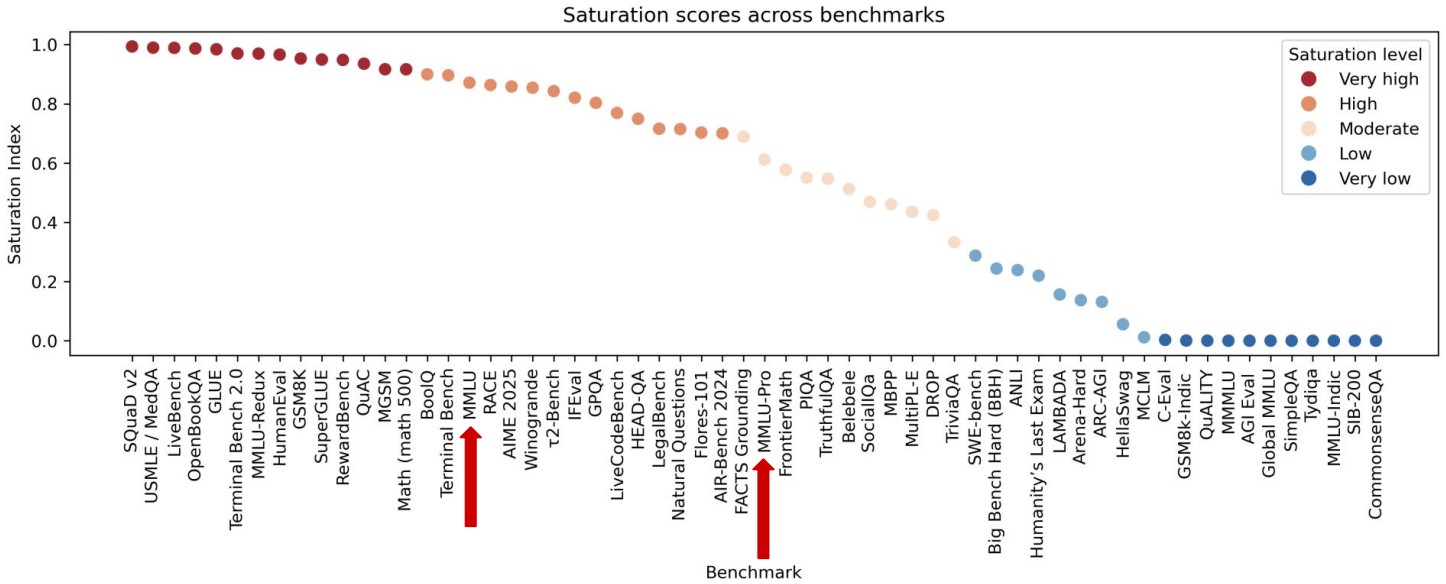
Benchmarks: Saturation



A benchmark is **saturated** "if the evaluated models cannot be reliably distinguished by their performance scores and any further improvements are not statistically distinguishable under the evaluation protocol."

[Akhtar et al., 2026]

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Benchmarks Today

Benchmarks Today: Humanity's Last Exam

https://lastexam.ai/

Classics

Question:



Here is a representation of a Roman inscription, originally found on a tombstone. Provide a translation for the Palmyrene script. A transliteration of the text is provided: RGYNᵓ BT ḤRY BR ᶜTᵓ ḤBL

Henry T
Merton College, Oxford

Ecology

Question:

Hummingbirds within Apodiformes uniquely have a bilaterally paired oval bone, a sesamoid embedded in the caudolateral portion of the expanded, cruciate aponeurosis of insertion of m. depressor caudae. How many paired tendons are supported by this sesamoid bone? Answer with a number.

Edward V
Massachusetts Institute of Technology

- Over 1,000 expert contributors!
- 2,500 challenging questions
- Tests structured academic problems

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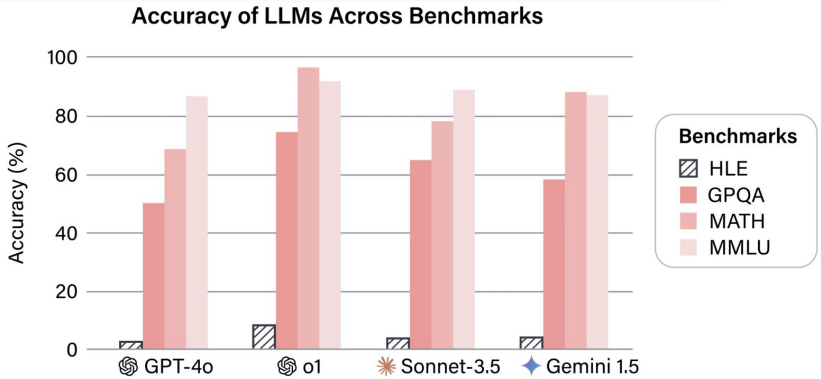
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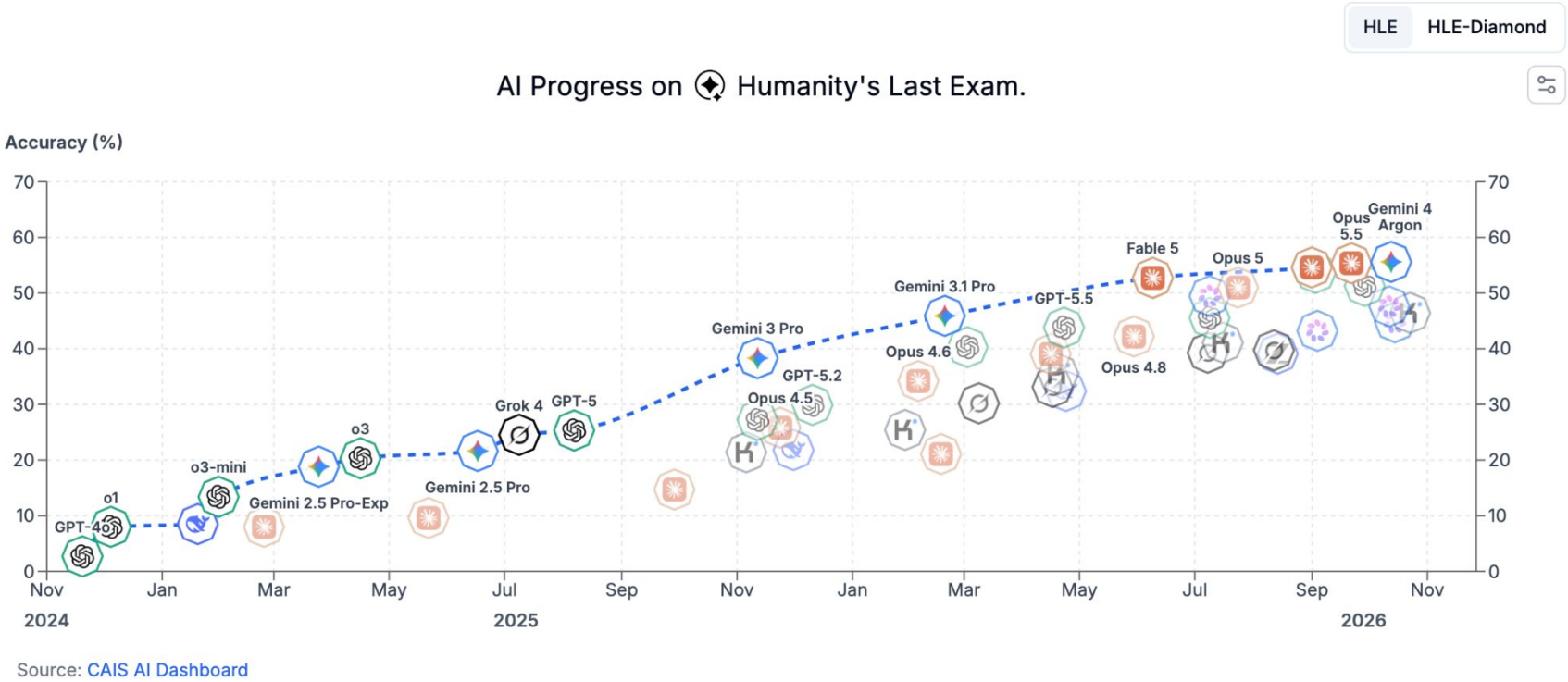
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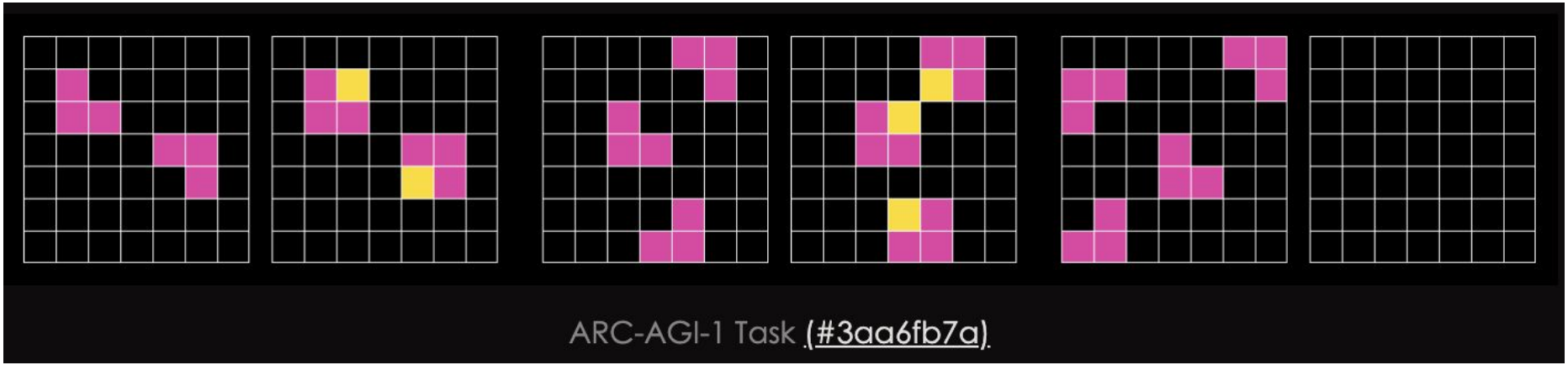


Benchmarks Today: Humanity's Last Exam



"HLE may be the last academic exam we need to give to models, but it is far from the last benchmark for AI" **Thoughts?**

Benchmarks Today: ARC-AGI Challenge

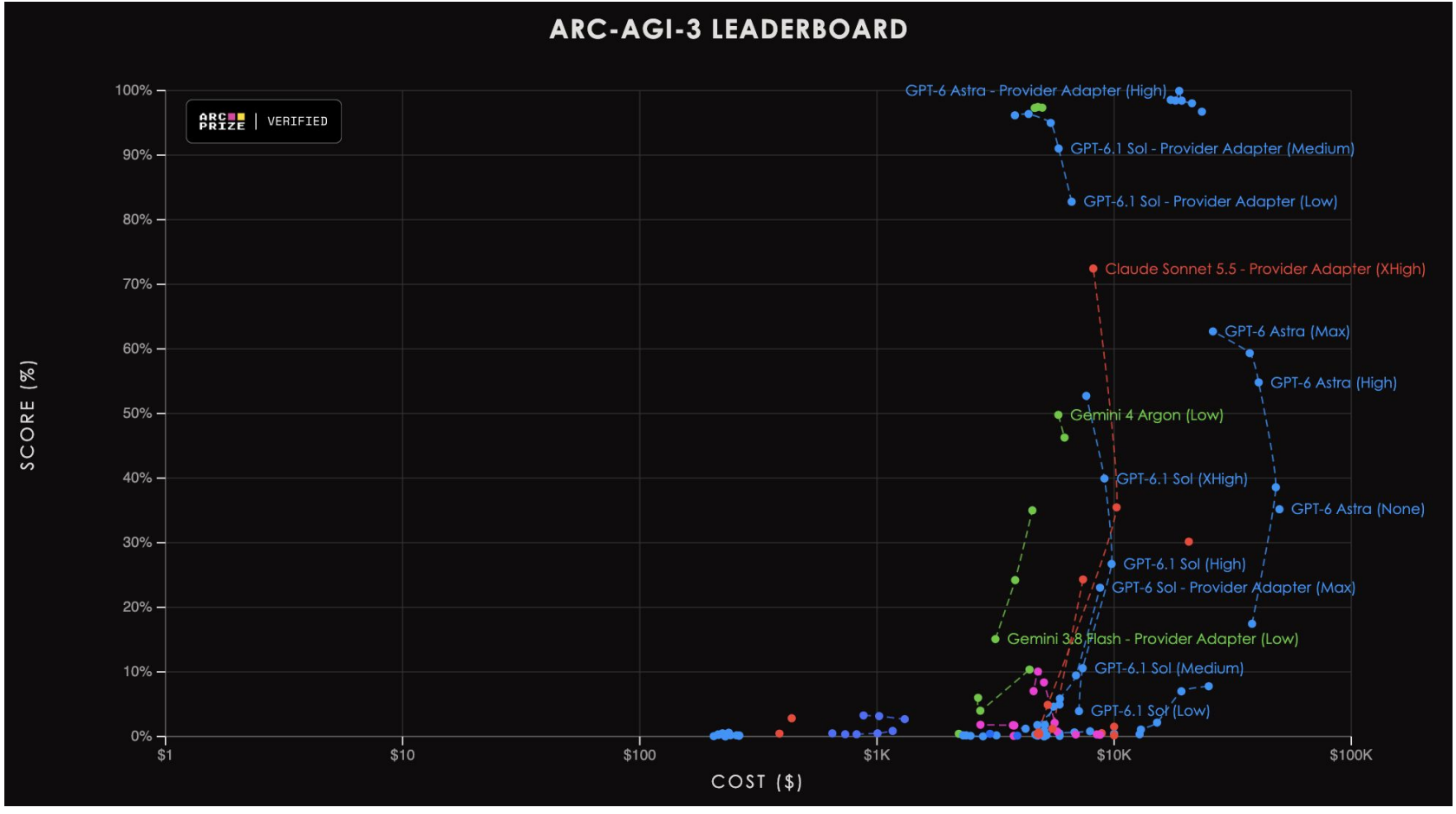


Abstraction and Reasoning Corpus (ARC-AGI-1)

ARC-AGI-2: https://arcprize.org/tasks/a6f40cea

ARC-AGI-3: https://arcprize.org/tasks/ls20

Benchmarks Today: ARC-AGI Challenge



Concluding Thoughts

Science of evaluation is lagging behind engineering progress

- Measuring understanding vs. pattern matching
- Preventing benchmark contamination and gaming
- Holistic evaluation of all aspects of LLM capabilities?
- Meta-evaluation:
  - Evaluation of evaluation benchmarks
  - Evaluation of evaluation metrics

TODOs:

- **Homework 3: Due 11/20**
- **Paper Discussion I**

Next Class: Paper Discussion I